

ECE 684 Natural Language Processing

Assignment 9 - Seq2Seq

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~~Matrix~~ We

The tokens = $\{a, b, c, d, e, \text{EOS}\}$

We have a -1 for the EOS and the matrix is used in the encoding step to transform the combined input and hidden state to a new hidden state. Based on this each row/column has either 0, 1, or -1s.

$$W_{ei} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Matrix W_{hh} : This has a 6x6 dimension and defines the transition between hidden states in the RNN. This updates the hidden states in the RNN. This updates the hidden state at each time step. Therefore the entries of 0's and 1's suggest a simple straightforward mechanism.

$$W_{hh} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Matrix W_o : This has a 2×6 dimension that produces the output by combining specific components of the hidden state. The hidden matrix distill the information into the necessary output. We have a -1 to flag the condition in hidden state that is relating to detecting the EoS, i.e., end of sequence.

$$W_o = \begin{bmatrix} 1 & 1 & -0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$